Activity 6 Report: Database Backup, Restoration, and Recovery

COMP 018 Database Administration | Project tested: BoardTrack (Boarding House and Rental Management) (Node.js + PostgreSQL) | Date of testing: 6 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	BoardTrack (Boarding House and Rental Management)
Owning group	Group 11 (BoardTrack)
Repository	https://github.com/karlrobertc-art/BoardTrack
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a6_backup.txt, a6_restore.txt, a6_verify.txt, a6_access.txt, a6_recovery.txt

Run on 6 November 2026 at 23:44 (Asia/Manila), after the application tests. The original database and the dump were never modified.

2. Backup plan

Item	Plan
Objects	Whole database boardtrack_db: 8 tables, 1 view, 7 sequences, 2 triggers, 2 extensions, constraints and grants
Method	Logical backup with pg_dump custom format (-Fc)
Account	The project defines no backup role, so the dump was taken as postgres. Recommend a role with pg_read_all_data
Files outside the database	The .env file (database URL and session secret) is not in the repository and must be kept separately. The app stores no uploaded files
Schedule (real deployment)	Daily full dump plus a dump before each change to the schema
Retention	7 daily and 4 weekly copies on a second machine
RPO and RTO	RPO 24 hours, RTO 1 hour
Roles	Roles are cluster-wide and not in a database dump. Keep a roles-only dump (pg_dumpall --roles-only)
Sensitive content	The dump holds tenant names, phone numbers, valid ID numbers, payment records and password hashes, so it must be stored with restricted access

3. Backup creation

```
pg_dump -U postgres -h localhost -Fc -d boardtrack_db -f boardtrack_db_activity6.dump
```

Item	Value

File	boardtrack_db_activity6.dump
Size	29,089 bytes
SHA-256	F733C9A01848F25611E17CC86C2F1E9A6B8B0498C1E32180379145E386A3F8E7
Archive	custom format, gzip, 79 TOC entries, 8 TABLE DATA entries, dumped by pg_dump 18.6
Errors	None, exit code 0

4. Restoration and verification

```
createdb -U postgres -h localhost boardtrack_db_restore
pg_restore -U postgres -h localhost -d boardtrack_db_restore boardtrack_db_activity6.dump
```

Restore succeeded: exit code 0 in 0.39 seconds, no warnings or errors.

Check	Original	Restored	Match
Tables, views, sequences	8, 1, 7	8, 1, 7	Yes
Row counts (app_user, lease, payment, property, room, schema_migrations, tenant, utility_bill)	3, 4, 7, 2, 6, 1, 4, 4	same	Yes
Constraints: check, foreign key, not null, primary key, unique, exclusion	15, 6, 36, 8, 3, 1	same	Yes
Table grants	63	63	Yes
Triggers	trg_lease_sync_room, trg_room_status_guard	same	Yes
Extensions	btree_gist, pgcrypto	same	Yes
Overlapping ACTIVE lease rejected (exclusion constraint)	rejected	rejected	Yes
Room status guard raises its message	yes	yes	Yes
Demo login check with crypt() on the stored hash	1	1	Yes
Application connects to the restored database	not required	not tested	n/a

5. Access control after restore

BoardTrack defines no database roles, so there was nothing to lose in the dump: datacl is empty (defaults) in both databases and the object owner is postgres in both. The four roles created for the Activity 4 test hold no privileges on either database, because the sketch was applied only to a separate copy. The test confirms that roles are not carried by pg_dump.

6. Recovery simulation

Step	Action	Result
Before	Checksum of payment in the restored copy	7 rows, md5 f41104ce34dcc2914fbcbcb10a61b8a5
Incident	DELETE of one payment row in the restored copy	DELETE 1, 6 rows, md5 b98f991bd602d9064c6f2a828a48b5c7
Recovery	createdb ..._recovery and pg_restore of	Success, exit code 0

	the same dump	
Compare	Checksum of payment in the recovery database	7 rows, same md5 f41104ce…
Original untouched	Checksum in the original database	7 rows, same md5

Outcome: Recovered. A full restore into a fresh database was chosen because it leaves the damaged copy untouched for comparison. Data written after the last backup would be lost, which is the RPO.

7. Evaluation and recommendations

- Daily dumps are enough for a small boarding house. Payments are the most important records, so a shorter RPO would need WAL archiving.
- The restore took 0.39 seconds, far inside the 1-hour RTO. Measure again as data grows.
- The target server needs the btree_gist and pgcrypto extensions (standard with PostgreSQL), otherwise the restore of the schema fails.
- Keep a roles-only dump and the role script with each backup once the project defines roles.
- Create a dedicated backup role and protect the dump files because they contain personal data and password hashes.

8. Limitations and errors

- Logical dumps capture one point in time only.
- The restore used the administrator account, so restored objects belong to postgres, the same as the original.
- No errors or warnings occurred.
- The application was not pointed at the restored database. The test databases stay until the evidence is captured and are then dropped.

9. Feedback sent to owners

Sent on 6 November 2026: the dump and restore worked without warnings, and the constraints, triggers and extensions all came back and still enforce the rules. The project has no backup role and no database roles.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own server, and every result in this report comes from the saved output files.